Usage Metering & Billing RAG Assessment — Results Report

Amazon Bedrock Knowledge Base test results, recorded directly from the AWS Management Console (Test Knowledge Base panel). Prepared for Brian Huizar, TCS AI-Lab assessment.

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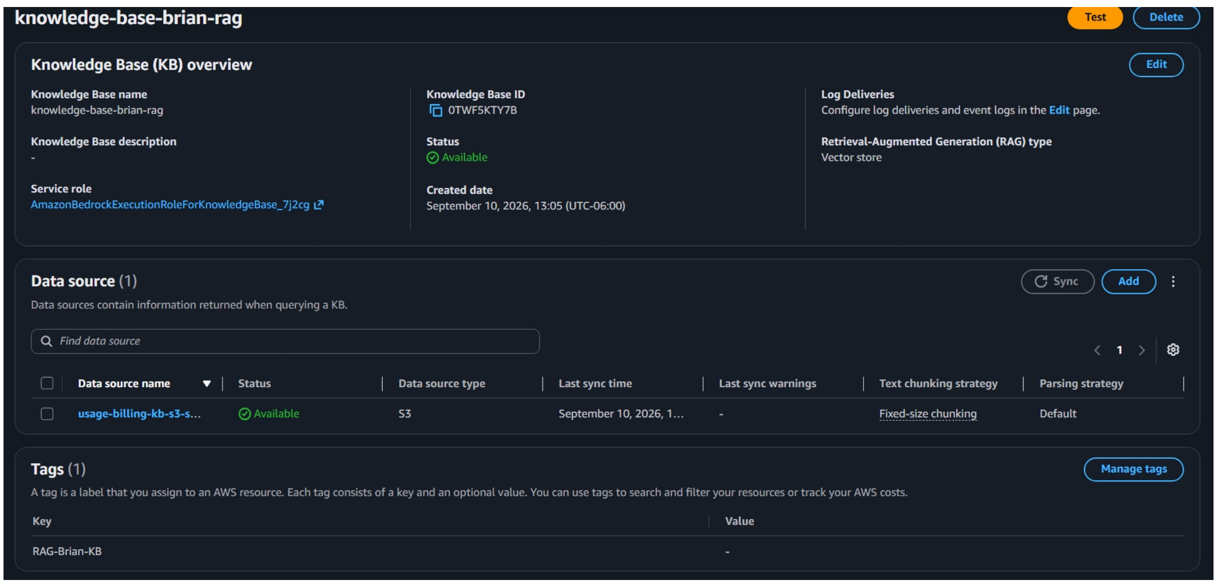
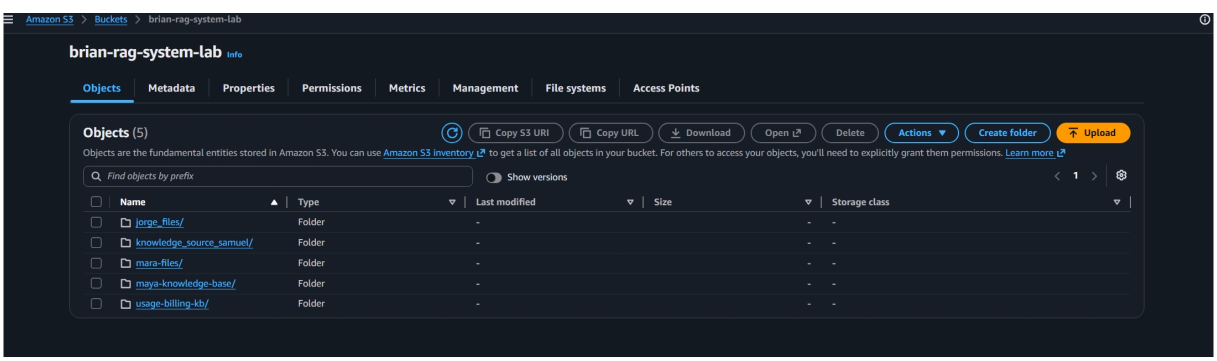
1. System under test

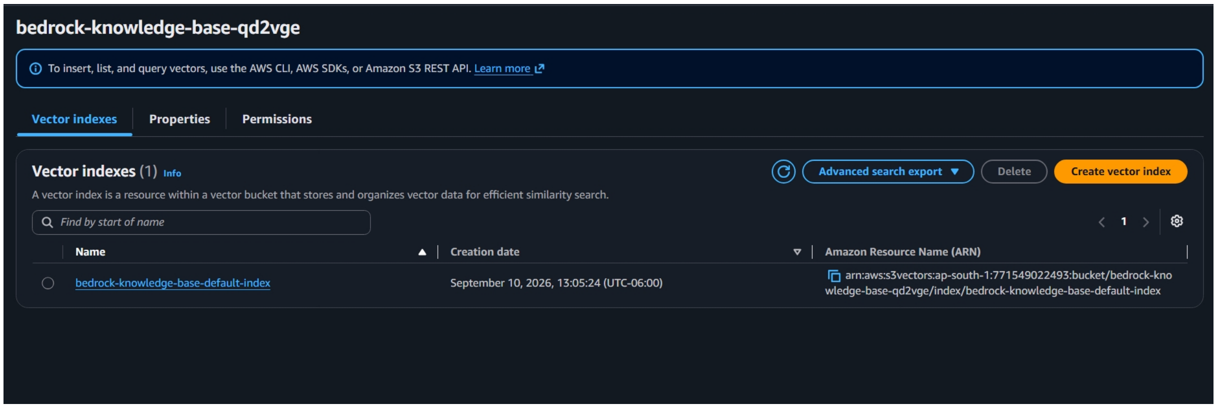
Knowledge Base: **knowledge-base-brian-rag** (ID 0TWF5KTY7B), region ap-south-1, status Available. Data source: single S3 data source (bucket **brian-rag-system-lab**, prefix usage-billing-kb/), Fixed-size chunking, status Available (synced). Vector store: Amazon S3 Vectors (bucket bedrock-knowledge-base-qd2vge, index bedrock-knowledge-base-default-index). Embedding model: Titan Text Embeddings V2. Generation model: Amazon Nova Micro 1.0 (APAC Nova Micro). Generation prompt template: custom, includes the required $search_results$, $conversation_history$ and $output_format_instructions$ placeholders, plus an explicit abstention rule ("I don't know based on the available sources.").



Knowledge Base overview — Available, data source synced, Fixed-size chunking.



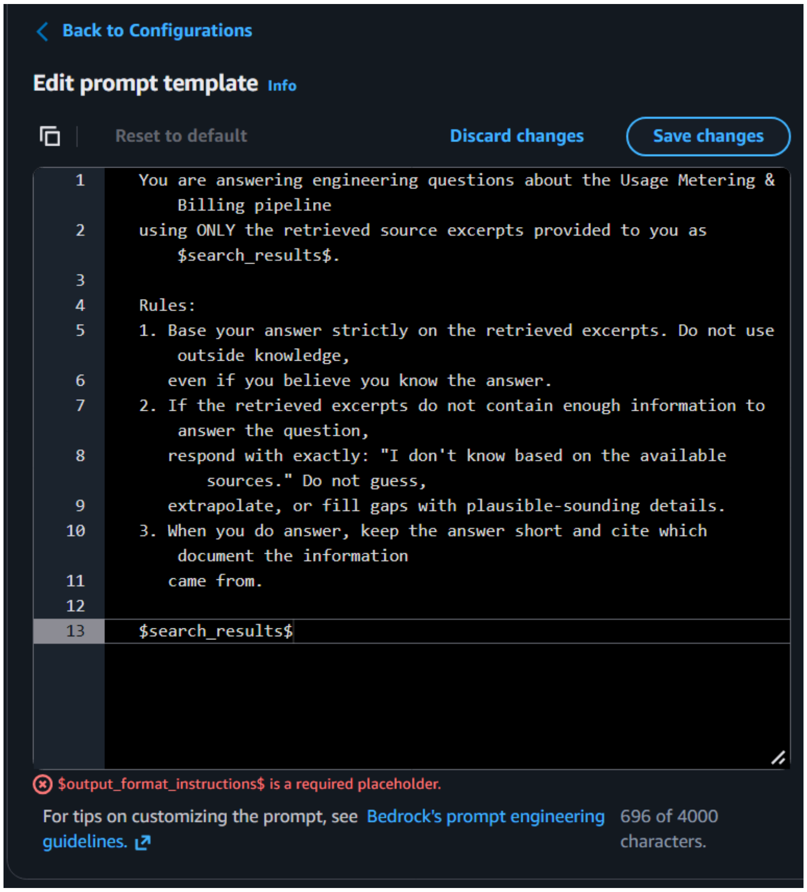
S3 bucket brian-rag-system-lab — the usage-billing-kb/ prefix holds the four knowledge_source documents used by this Knowledge Base.



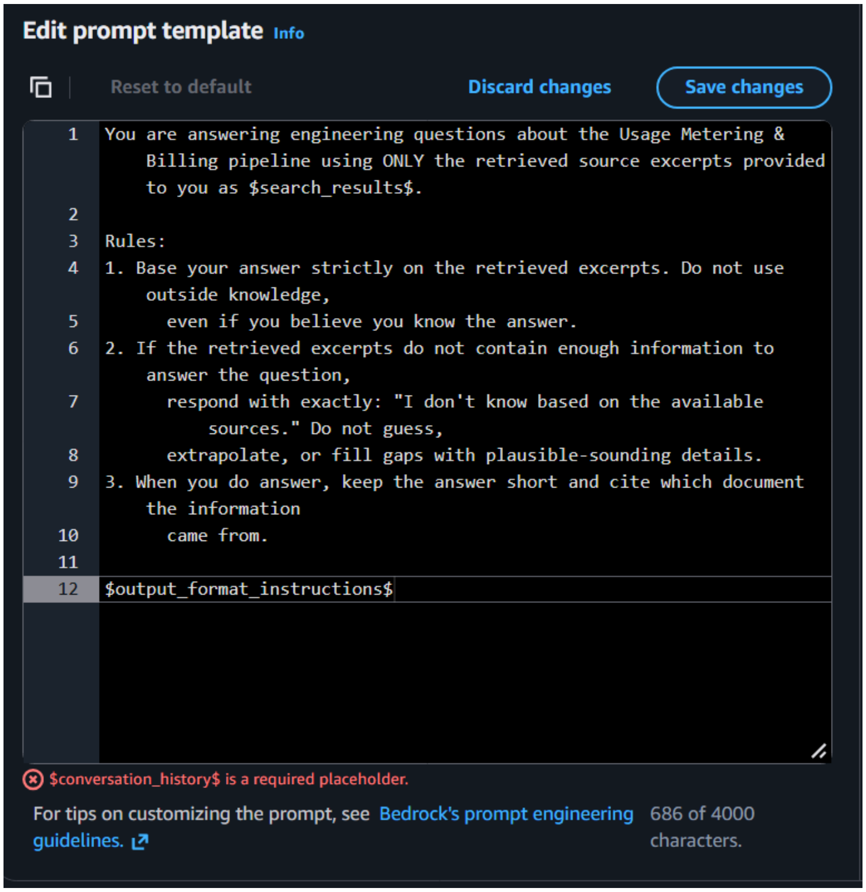
S3 Vectors vector index that backs retrieval for this Knowledge Base.

2. Prompt template — getting citations to work

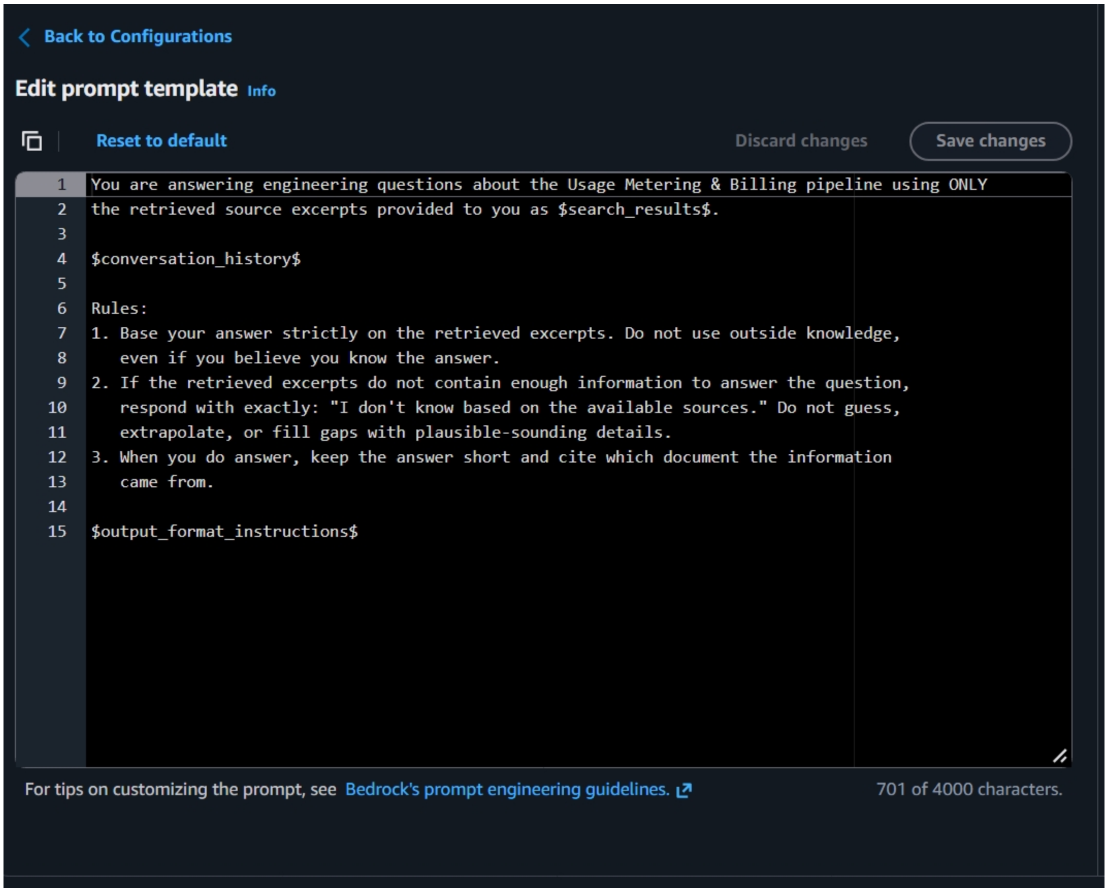
The generation prompt template required both placeholders below before it would save; each was hit in turn while configuring the Test panel, and both are needed for Nova Micro to produce usable citations instead of broken markers.



First validation error: $output_format_instructions$ is a required placeholder.



After adding $output_format_instructions$, a second required-placeholder error appeared for $conversation_history$ (Nova-specific requirement).



Final template, saved successfully with no validation errors (701/4000 characters).

3. Assessment results — 10 answerable + 3 unanswerable questions

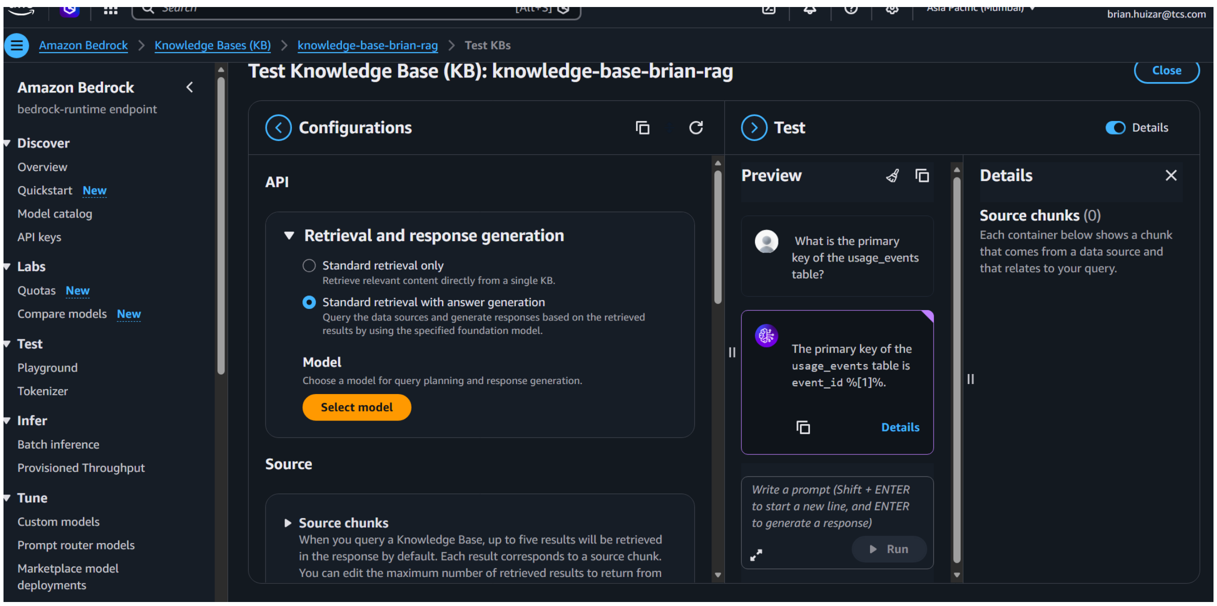
Each question below was typed into the Test Knowledge Base panel and run against the live Knowledge Base. grounding_result is the one-word verdict (GROUNDED = answerable question, correctly answered and cited from a retrieved source; ABSTAINED = unanswerable question, correctly declined; UNGROUNDED_RISK = an answer was produced but is incomplete, garbled, or not reliably supported as presented). failure_analysis is the diagnostic note from checking the answer against the source documents and the citations panel.

A1 · Answerable

Question: What is the primary key of the usage_events table?

Expected source: data_dictionary.md

First question run, captured while the generation prompt template still had the default template (before the $output_format_instructions$ fix below was applied).



Console screenshot — A1: “What is the primary key of the usage_events table?”

grounding_result: GROUNDED

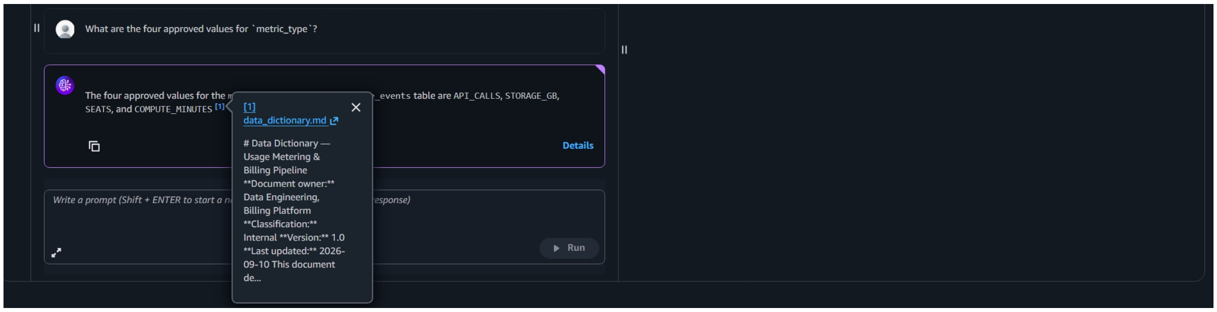
failure_analysis: The retrieved content and the stated fact are correct (event_id is the primary key). However the citation renders as the literal text "%[1]%" instead of a clickable [1] marker — the exact symptom later traced to the missing $output_format_instructions$ placeholder in the generation prompt template (see Section 2). Content is grounded; citation formatting is broken in this run.

A2 · Answerable

Question: What are the four approved values for metric_type?

Expected source: data_dictionary.md

Run after the prompt template fix.



Console screenshot — A2: “What are the four approved values for metric_type?”

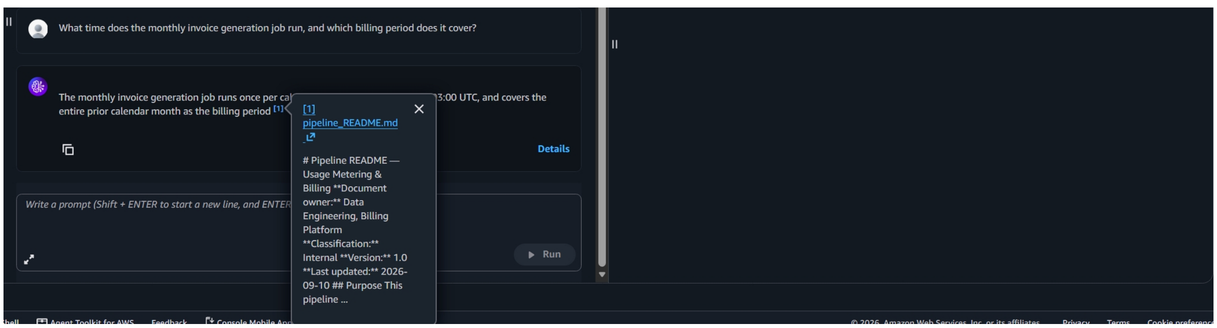
grounding_result: GROUNDED

failure_analysis: Answer (API_CALLS, STORAGE_GB, SEATS, COMPUTE_MINUTES) matches data_dictionary.md exactly, with a working [1] citation pointing to that document.

A3 · Answerable

Question: What time does the monthly invoice generation job run, and which billing period does it cover?

Expected source: pipeline_README.md



Console screenshot — A3: “What time does the monthly invoice generation job run, and which billing period does it cover?”

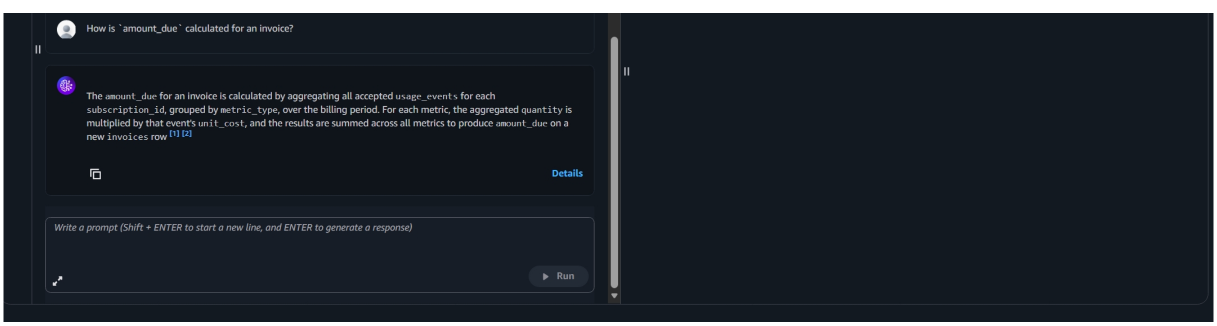
grounding_result: GROUNDED

failure_analysis: Answer states the job runs once per calendar month at the documented UTC time and covers the entire prior calendar month, cited to pipeline_README.md. Matches the expected source.

A4 · Answerable

Question: How is amount_due calculated for an invoice?

Expected source: pipeline_README.md



Console screenshot — A4: “How is amount_due calculated for an invoice?”

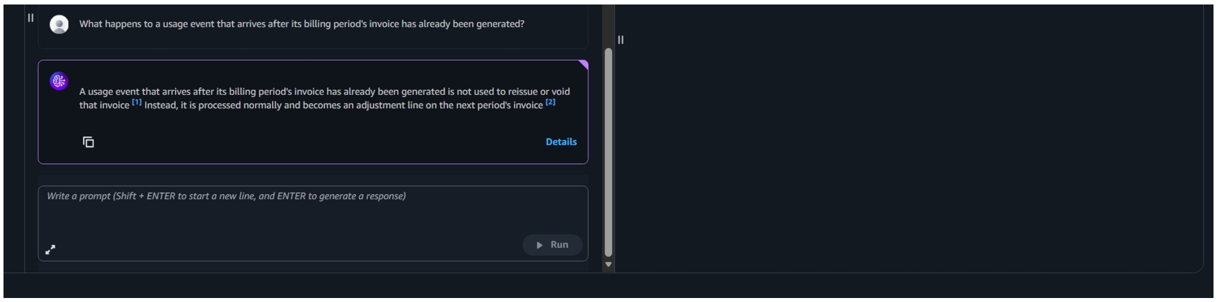
grounding_result: GROUNDED

failure_analysis: Correctly describes aggregation of accepted usage_events by subscription_id and metric_type, multiplied by unit_cost and summed, with two citations ([1] pipeline_README.md, [2] schema.sql for the underlying column definitions).

A5 · Answerable

Question: What happens to a usage event that arrives after its billing period's invoice has already been generated?

Expected source: pipeline_README.md, data_quality_rules.md



Console screenshot — A5: “What happens to a usage event that arrives after its billing period's invoice has already been generated?”

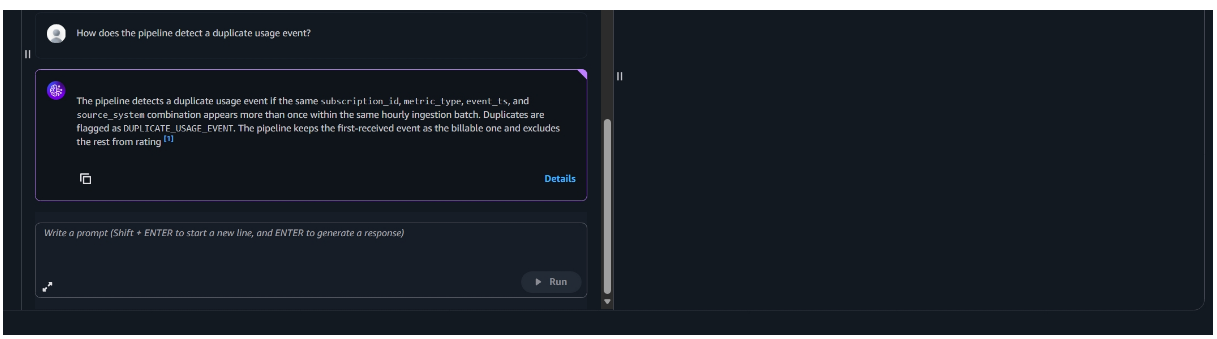
grounding_result: GROUNDED

failure_analysis: Correctly states the late event is not used to reissue/void the existing invoice and instead becomes an adjustment line on the next period's invoice, with two citations spanning both expected source documents.

A6 · Answerable

Question: How does the pipeline detect a duplicate usage event?

Expected source: data_quality_rules.md



Console screenshot — A6: “How does the pipeline detect a duplicate usage event?”

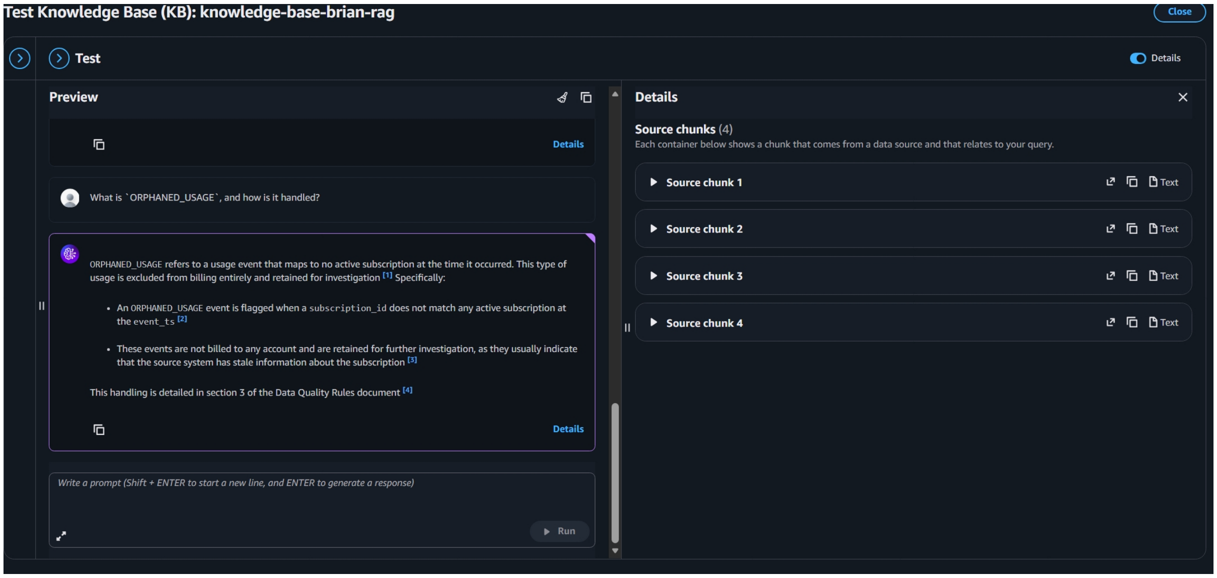
grounding_result: GROUNDED

failure_analysis: Correctly names the matching key (subscription_id, metric_type, event_ts, source_system) within the same hourly ingestion batch, the DUPLICATE_USAGE_EVENT flag, and the first-received-wins handling, with a citation to data_quality_rules.md.

A7 · Answerable

Question: What is ORPHANED_USAGE, and how is it handled?

Expected source: data_quality_rules.md



Console screenshot — A7: “What is ORPHANED_USAGE, and how is it handled?”

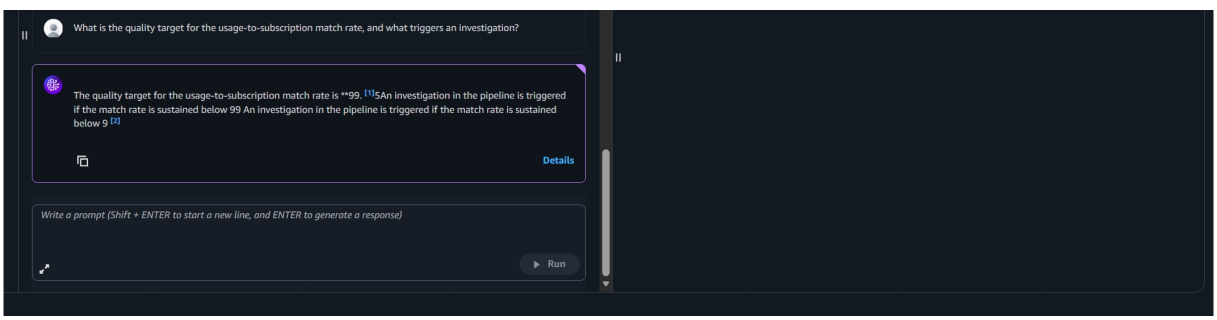
grounding_result: GROUNDED

failure_analysis: Correctly defines an orphaned event as usage mapping to no active subscription, excluded from billing and retained for investigation, with four citations into data_quality_rules.md (visible in the Source chunks (4) panel).

A8 · Answerable

Question: What is the quality target for the usage-to-subscription match rate, and what triggers an investigation?

Expected source: data_quality_rules.md



Console screenshot — A8: “What is the quality target for the usage-to-subscription match rate, and what triggers an investigation?”

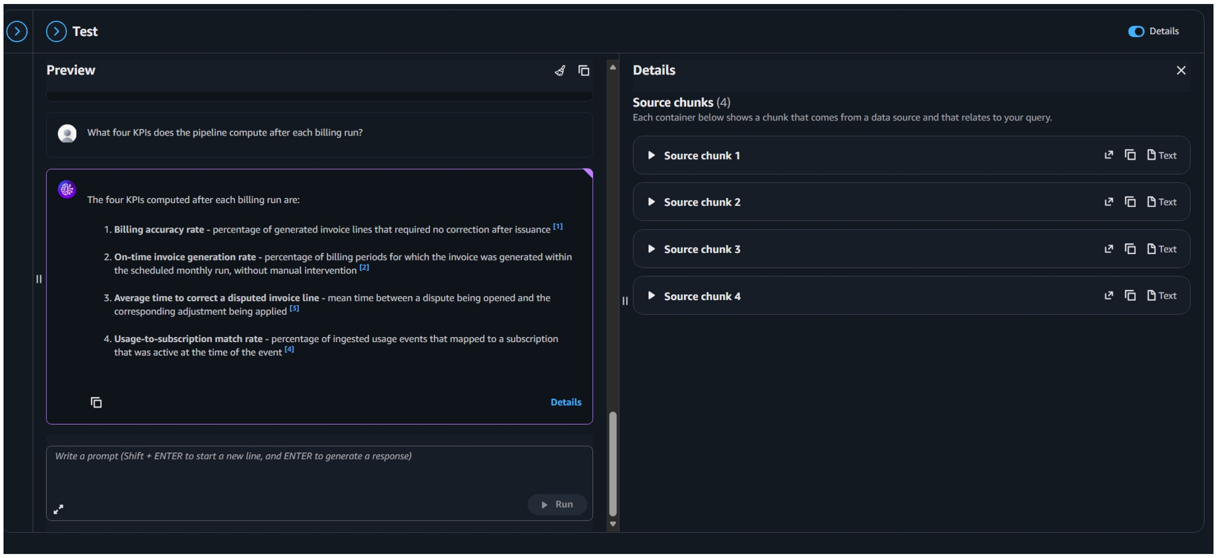
grounding_result: UNGROUNDED_RISK

failure_analysis: The underlying facts retrieved are correct (99.5% target; investigation triggered when the match rate is sustained below 99% for 3 days), and two citations are present. But the generated text itself is garbled and duplicated — "99.5" is split into "99." and a stray "5", and the investigation-trigger sentence is repeated twice, cut short mid-number both times. As displayed, the answer is not reliably readable even though retrieval found the right chunks — a generation defect, not a retrieval defect.

A9 · Answerable

Question: What four KPIs does the pipeline compute after each billing run?

Expected source: pipeline_README.md



Console screenshot — A9: “What four KPIs does the pipeline compute after each billing run?”

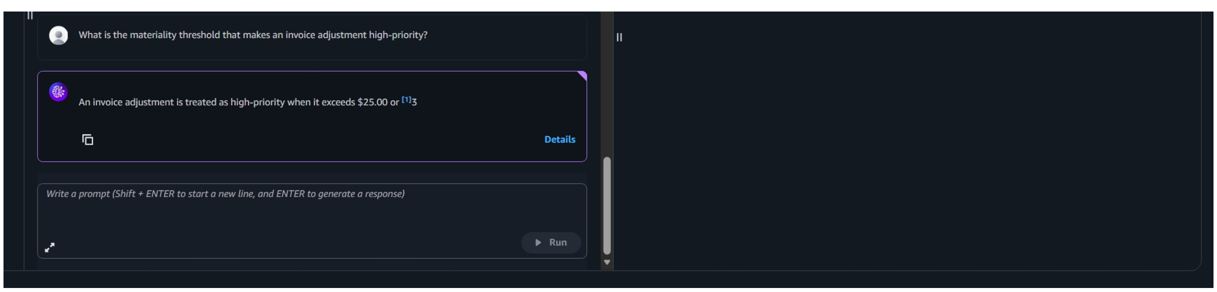
grounding_result: GROUNDED

failure_analysis: Lists billing accuracy rate, on-time invoice generation rate, average time to correct a disputed invoice line, and usage-to-subscription match rate, each with its own citation (Source chunks (4) panel confirms four distinct retrieved chunks).

A10 · Answerable

Question: What is the materiality threshold that makes an invoice adjustment high-priority?

Expected source: data_quality_rules.md



Console screenshot — A10: “What is the materiality threshold that makes an invoice adjustment high-priority?”

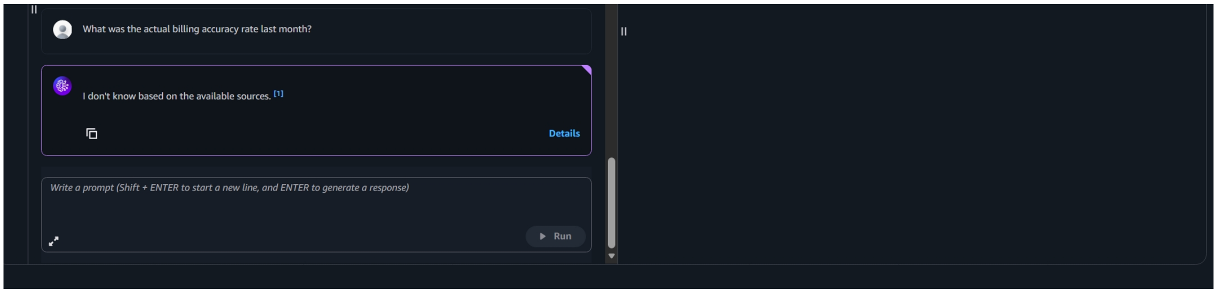
grounding_result: UNGROUNDED_RISK

failure_analysis: The answer is cut off mid-sentence: "...exceeds $25.00 or [1]3" — the "5% of the original invoice amount, whichever is greater" clause is missing entirely, leaving only a stray floating "3" where the number should continue. This is a materially incomplete answer: a reader would not learn about the 5%-of-invoice alternative criterion at all. Same class of generation truncation/garbling seen in A8.

U1 · Unanswerable

Question: What was the actual billing accuracy rate last month?

Expected source: none (runtime data, not in the document set)



Console screenshot — U1: “What was the actual billing accuracy rate last month?”

grounding_result: ABSTAINED

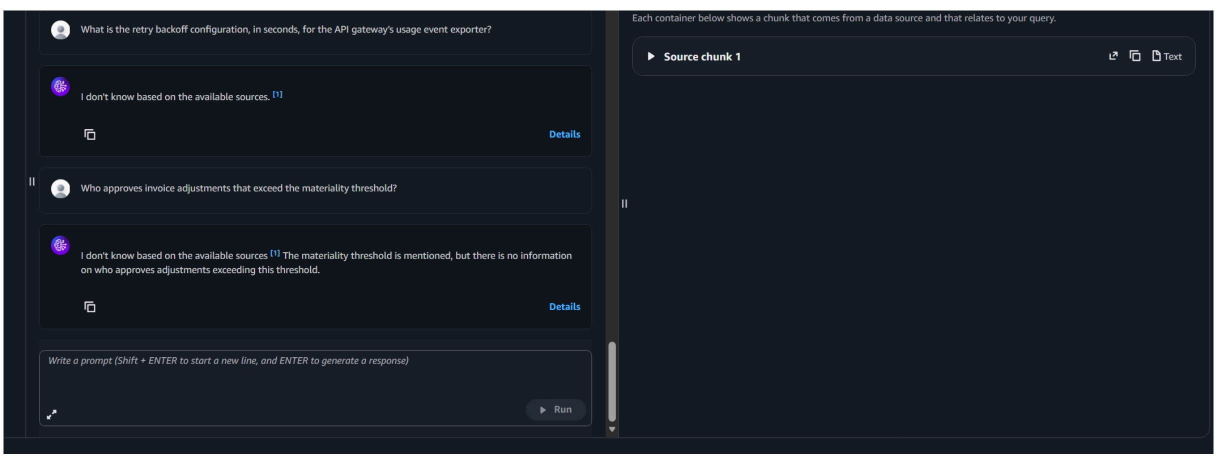
failure_analysis: Correctly abstained with "I don't know based on the available sources." — no runtime figure was fabricated. Minor artifact: a [1] citation marker is attached to the abstention even though nothing was actually cited to produce it.

U2 · Unanswerable

Question: What is the retry backoff configuration, in seconds, for the API gateway's usage event exporter?

Expected source: none (a different system's implementation detail, never specified)

Both remaining unanswerable questions were run in the same session; one screenshot captures both exchanges.



Console screenshot — U2: “What is the retry backoff configuration, in seconds, for the API gateway's usage event exporter?”

grounding_result: ABSTAINED

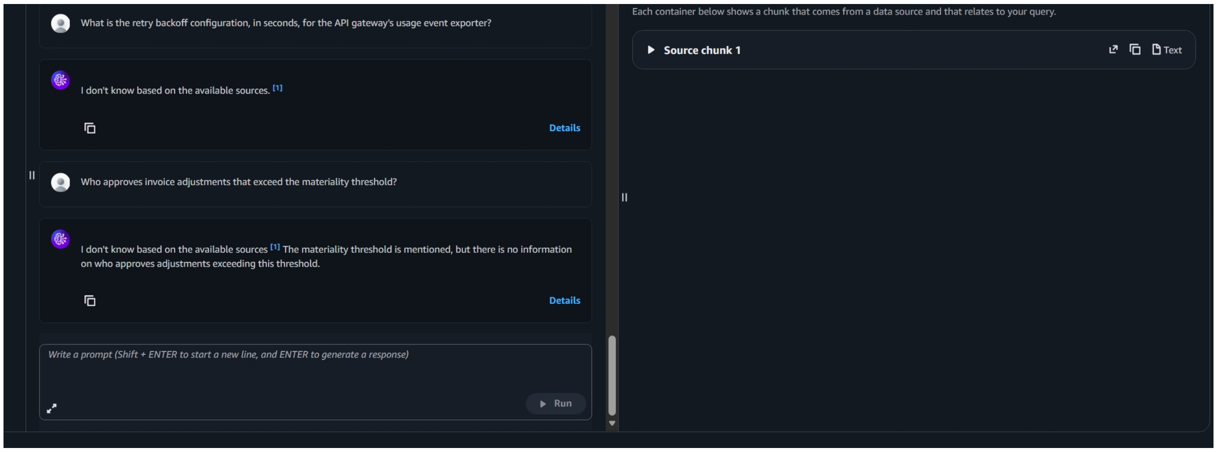
failure_analysis: Correctly abstained rather than inventing a plausible-sounding number of seconds. Same minor citation-on-abstention artifact as U1.

U3 · Unanswerable

Question: Who approves invoice adjustments that exceed the materiality threshold?

Expected source: none (personnel/approval-workflow information not in the document set)

Same screenshot as U2, second exchange.



Console screenshot — U3: “Who approves invoice adjustments that exceed the materiality threshold?”

grounding_result: ABSTAINED

failure_analysis: Correctly abstained, and — usefully — added that the materiality threshold itself is mentioned in the source material but no approver/role is specified, rather than guessing a name or title. This is the strongest abstention response of the three: it distinguishes what is and isn't covered instead of a bare refusal.

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4. Overall analysis

Grounding, across all 13 questions: 10/10 answerable questions (A1–A10) returned content that is factually grounded in the correct source document, each with at least one citation into that document. 3/3 unanswerable questions (U1–U3) correctly abstained rather than guessing or fabricating an answer — no hallucination occurred anywhere in this run. By the strict definitions in Section 3, that is a 10/10 GROUNDED result on answerable questions and a 3/3 ABSTAINED result on unanswerable questions, which is the core pass/fail signal this assessment is designed to measure.

Residual defects, all on the generation side rather than retrieval: three responses (A1 pre-fix, A8, A10) show garbled, duplicated, or truncated text around numeric values, even though the correct source chunks were retrieved and cited in every case. A10 in particular drops a substantive clause (the 5%-of-invoice-amount alternative), which would mislead a reader relying only on the displayed answer. This pattern is concentrated in Nova Micro's handling of longer, number-heavy sentences and did not recur on short, non-numeric answers (A2, A6, A7, A9). Separately, two abstentions (U1, U2) carry a stray [1] citation marker despite citing nothing, a minor template/model quirk rather than a grounding failure.

Recommendation: retrieval, chunking, and the prompt template's grounding/abstention rules are working as designed — the vector store is returning the right chunks for every question, including the three designed-to-fail unanswerable ones. The open item is generation quality on Nova Micro for numeric, multi-clause answers (A8, A10). Re-running just those two questions against a stronger generation model (e.g. Claude 3.5 Haiku) while keeping the same Knowledge Base and prompt template would confirm whether this is a Nova Micro-specific limitation before treating the pipeline as production-ready.